

Lecture 6b

EE-215 Electronic Devices and Circuits

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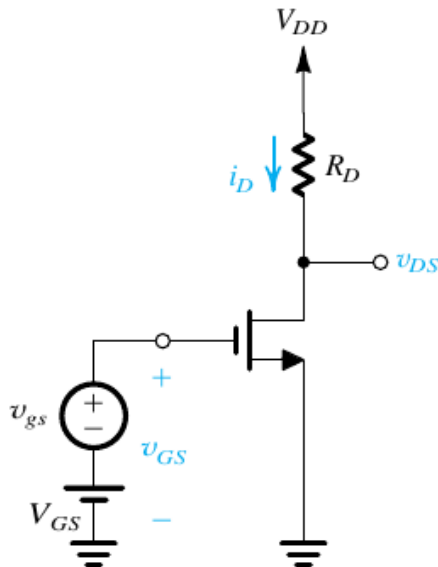
Small-Signal Operation and Models

- Thus for MOSFET amplifier, linear amplification can be obtained by
 - biasing the MOSFET to operate in the saturation region
 - and by keeping the input signal small

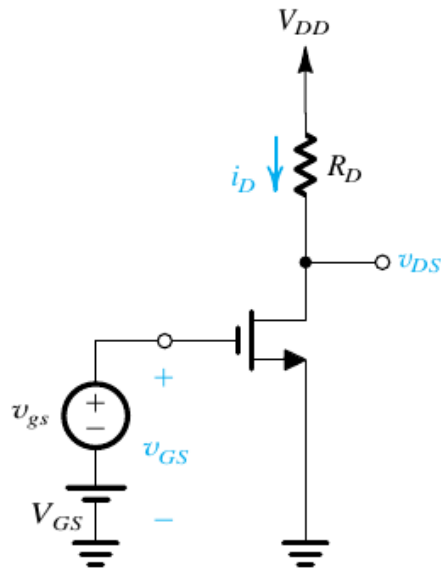
The DC Bias Point

- for the conceptual amplifier circuit shown in fig
 - here the MOSFET is biased by applying a dc voltage V_{GS}
 - and the input signal to be amplified v_{gs} is superimposed on the dc bias voltage V_{GS}
 - the output voltage is v_{DS}
 - the dc bias current I_D can be determined by setting ac signal $v_{gs} = 0$

$$\blacksquare = \Rightarrow I_D = \frac{1}{2}k_n' \frac{W}{L}(V_{GS} - V_t)^2 = \frac{1}{2}k_n(V_{OV})^2 \text{ if } \lambda = 0$$



- the dc voltage at the drain can be given as (using KVL)
 - $V_{DD} = I_D R_D + V_{DS}$
 - $V_{DS} = V_{DD} - I_D R_D$
 - to ensure saturation region of operation
 - $V_{DS} > V_{OV}$
 - As the total voltage at the drain will have a signal component superimposed on V_{DS}
 - $\Rightarrow V_{DS}$ has to be sufficiently greater than V_{OV} to
 - allow for the required signal swing



The Signal Current in the Drain Terminal

- Now if v_{gs} is applied, the total instantaneous gate-to-source voltage is

- $v_{GS} = V_{GS} + v_{gs}$

- and the total instantaneous drain current i_D is

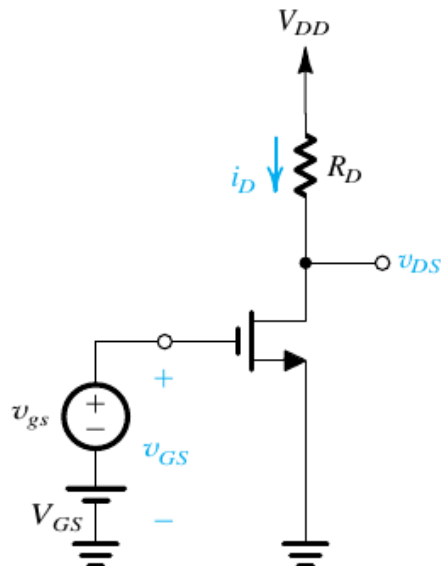
- $i_D = \frac{1}{2}k_n' \frac{W}{L} (v_{GS} - V_t)^2$

- $i_D = \frac{1}{2}k_n (v_{GS} - V_t)^2$

- $i_D = \frac{1}{2}k_n (V_{GS} + v_{gs} - V_t)^2$

- $i_D = \frac{1}{2}k_n [(V_{GS} - V_t) + v_{gs}]^2$

- $i_D = \frac{1}{2}k_n [(V_{GS} - V_t)^2 + 2(V_{GS} - V_t)v_{gs} + v_{gs}^2]$



- $i_D = \frac{1}{2}k_n (V_{GS} - V_t)^2 + k_n (V_{GS} - V_t)v_{gs} + \frac{1}{2}k_n v_{gs}^2$

- $i_D = \frac{1}{2}k_n (V_{GS} - V_t)^2 + k_n (V_{GS} - V_t)v_{gs} + \frac{1}{2}k_n v_{gs}^2$

- the first term is the dc bias current i.e. $I_D = \frac{1}{2}k_n(V_{GS} - V_t)^2$
 - $= \Rightarrow i_D = I_D + k_n(V_{GS} - V_t)v_{gs} + \frac{1}{2}k_nv_{gs}^2$
- the 2nd term of R.H.S represent a current component
 - that is directly proportional to v_{gs}
- the 3rd term is a current component that is proportional
 - to the square of the input signal.
 - this 3rd term represents non-linear distortion and is undesirable
- this non-linear distortion introduced by MOSFET can be
 - reduced by keeping the input signal small
 - i.e. 3rd term < 2nd term $\Rightarrow \frac{1}{2}k_nv_{gs}^2 < k_n(V_{GS} - V_t)v_{gs}$
- $\frac{1}{2}k_nv_{gs}^2 < k_n(V_{GS} - V_t)v_{gs}$
 - $\Rightarrow \frac{1}{2}v_{gs}^2 < (V_{GS} - V_t)v_{gs}$
 - $\Rightarrow v_{gs} < 2(V_{GS} - V_t)$
 - or $v_{gs} < 2V_{OV}$
 - this is the small-signal condition
 - If this condition is satisfied, we can ignore the 3rd term in the expression of i_D
 - $\Rightarrow i_D \approx I_D + k_n(V_{GS} - V_t)v_{gs}$
 - or $i_D \approx I_D + i_d$
 - where $i_d = k_n(V_{GS} - V_t)v_{gs}$ or $i_d = g_mv_{gs}$, $g_m = k_n(V_{GS} - V_t)$
 - thus $g_m = k_n(V_{GS} - V_t) = k_nV_{OV} = \frac{i_d}{v_{gs}}$
 - this parameter that relates the i_d and v_{gs} is called the MOSFET transconductance
- Graphical interpretation of the small signal operation of the MOSFET amplifier is as shown in figure
 - Note that g_m is equal to the slope of the $i_D - v_{GS}$ characteristic at the bias point
 - i.e. $g_m = \left. \frac{\partial i_D}{\partial v_{GS}} \right|_{v_{GS}=V_{GS}}$
 - as $i_D = \frac{1}{2}k_n(v_{GS} - V_t)^2$
 - $g_m = \left. \frac{\partial}{\partial v_{GS}} \left\{ \frac{1}{2}k_n(v_{GS} - V_t)^2 \right\} \right|_{v_{GS}=V_{GS}}$
 - $g_m = \left. \frac{1}{2}k_n 2(v_{GS} - V_t) \cdot 1 \right|_{v_{GS}=V_{GS}}$
 - $g_m = k_n(V_{GS} - V_t)$

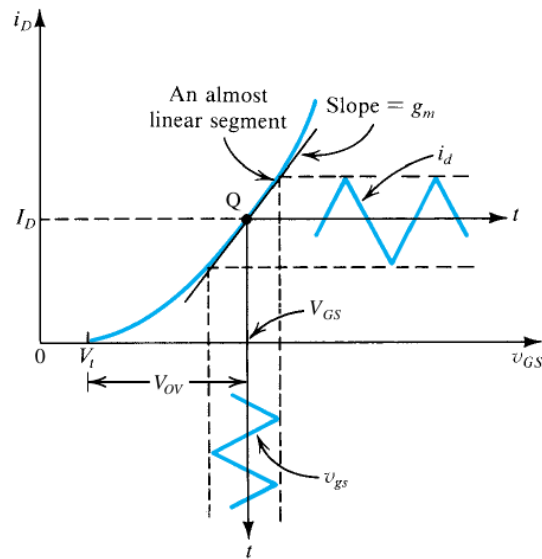


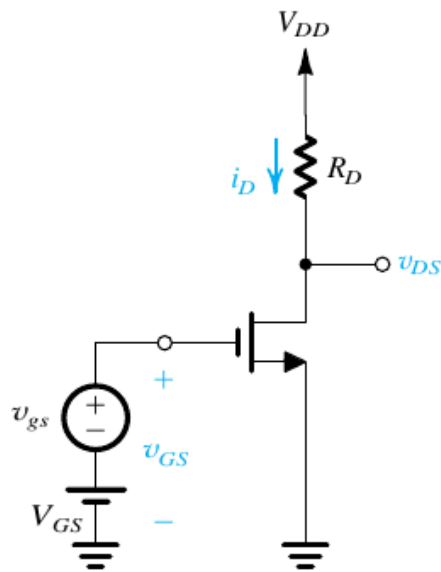
Figure 5.35 Small-signal operation of the MOSFET amplifier.

The Transconductance, g_m

- $g_m = k_n (V_{GS} - V_t) = k_n V_{OV}$
 - As
 - $I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2 = \frac{1}{2} k_n V_{OV}^2$
 - $\Rightarrow k_n V_{OV} = \frac{2I_D}{V_{OV}}$ or $V_{OV} = \sqrt{\frac{2I_D}{k_n}}$
 - $g_m = k_n V_{OV} = \frac{2I_D}{V_{OV}}$
 - $g_m = k_n V_{OV} = k_n \sqrt{\frac{2I_D}{k_n}} = \sqrt{k_n} \sqrt{2I_D} = \sqrt{2k_n'} \sqrt{\frac{W}{L}} \sqrt{I_D}$
 -
- $$g_m = k_n V_{OV} = \frac{2I_D}{V_{OV}} = \sqrt{2k_n'} \sqrt{\frac{W}{L}} \sqrt{I_D}$$

The Voltage Gain

- the total instantaneous drain voltage is
 - $v_{DS} = V_{DD} - R_D i_D$
 - under the small signal condition
 - $i_D = I_D + i_d$
 - $\Rightarrow v_{DS} = V_{DD} - R_D i_D = V_{DD} - R_D (I_D + i_d)$
 - $v_{DS} = V_{DD} - R_D I_D - R_D i_d$
 - $v_{DS} = V_{DS} - R_D i_d \because V_{DS} = V_{DD} - R_D I_D$
 - $\Rightarrow$ the signal component of v_{DS} is
 - $v_{ds} = -R_D i_d$
 - As $g_m = \frac{i_d}{v_{gs}} \Rightarrow i_d = g_m v_{gs}$



- $v_{ds} = -R_D i_d$, $i_d = g_m v_{gs}$

- $\Rightarrow v_{ds} = -R_D i_d = -R_D (g_m v_{gs})$

- $v_{ds} = -R_D g_m v_{gs}$ or $\frac{v_{ds}}{v_{gs}} = -R_D g_m$

- $\Rightarrow A_v = \frac{v_{ds}}{v_{gs}} = -R_D g_m$

- here the minus sign indicates that the output signal is 180° out of phase w.r.t. the input signal v_{gs}

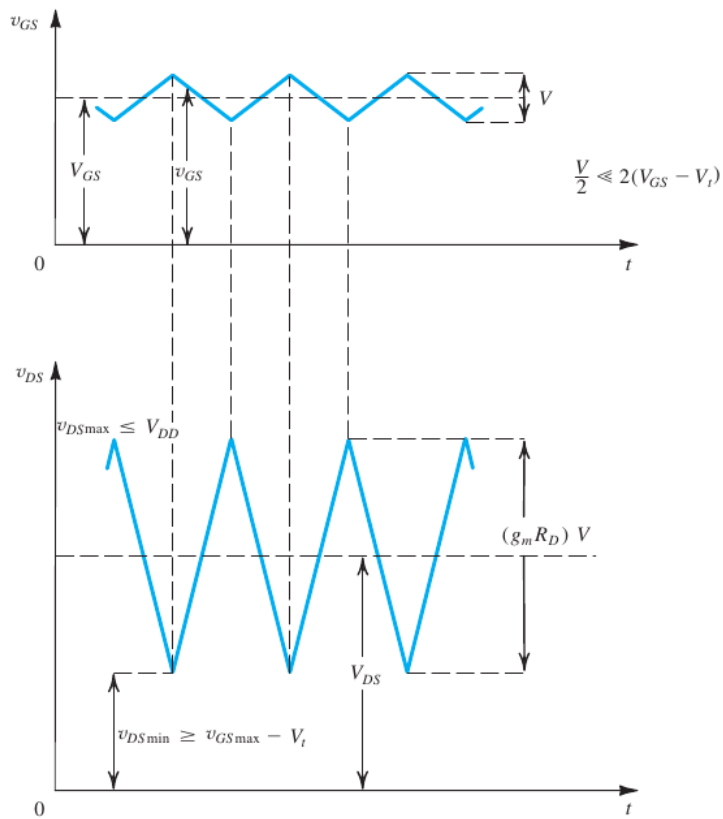


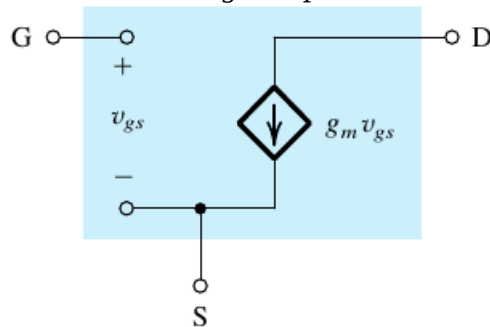
Figure 5.36 Total instantaneous voltages v_{GS} and v_{DS} for the circuit in Fig. 5.34.

Separating the DC Analysis and the Signal Analysis

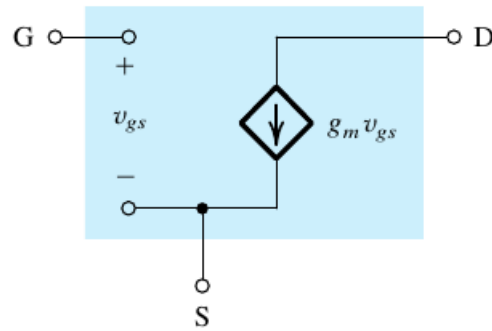
- thus under the small-signal approximation,
 - signal quantities are superimposed on dc quantities
 - $= \Rightarrow$ the total i_D equals the dc current I_D plus the signal current i_d
 - the total drain voltage v_{DS} equals $V_{DS} + v_{ds}$ and so on
 - thus the analysis and design can be greatly simplified by
 - separating dc calculations from
 - small-signal or ac calculations
 - first we perform dc analysis by suppressing all ac sources
 - once dc operating point is determined, we proceed with ac analysis
 - for ac analysis, we suppress all dc sources,
 - i.e. dc voltage source is replaced by a short circuit
 - a dc current source is replaced by an open circuit

Small-Signal Equivalent Circuit Models

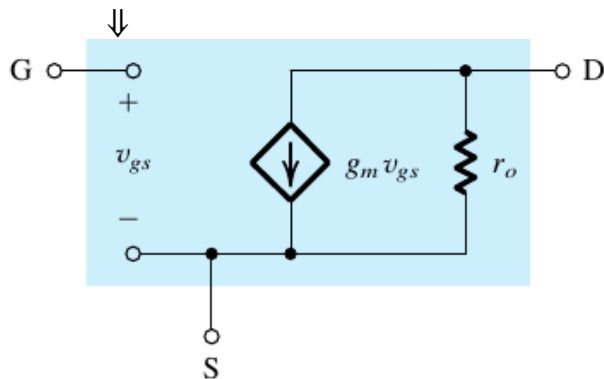
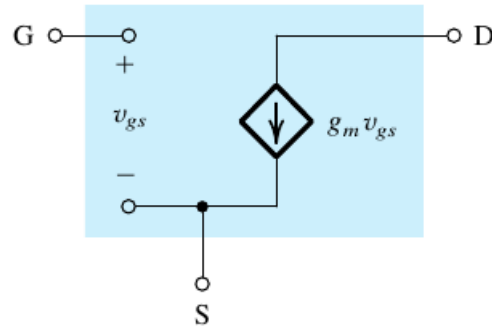
- we have already seen that under the small-signal condition
 - $i_d = g_m v_{gs}$ where $g_m = k_n (V_{GS} - V_t)$
 - $= \Rightarrow$ MOSFET behaves as a voltage controlled current source
 - it accepts a signal v_{gs} between gate and the source
 - and provides a current $g_m v_{gs}$ at the drain terminal
 - The input resistance at the gate is very high ideally infinite
 - the output resistance is also infinite (if $\lambda = 0$)
 - thus the small-signal operation of the MOSFET can be represented by figure



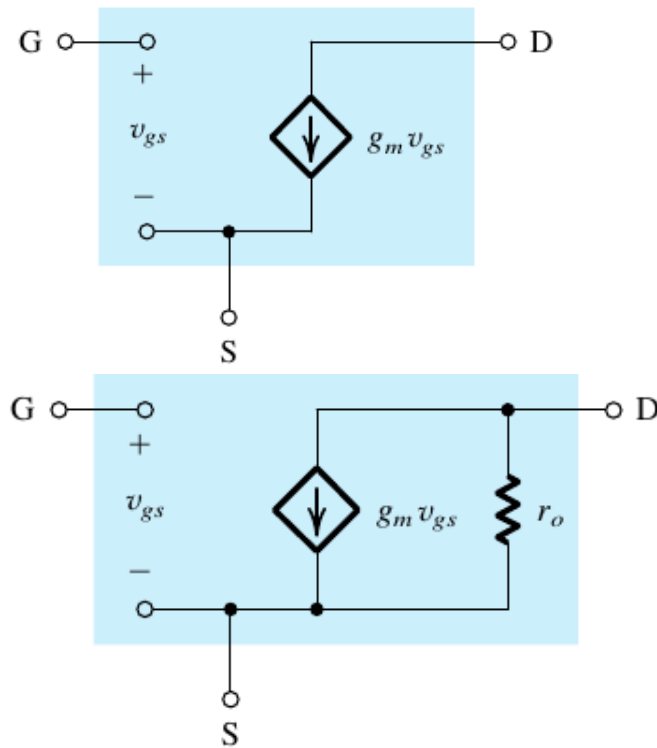
- this circuit shown in figure is called a small-signal model
 - this particular model is called the hybrid- π model
 - Thus in the analysis of a MOSFET amplifier circuit, we
 - first perform DC analysis
 - Once dc analysis is done,
 - replace the MOSFET by its small-signal model,
 - suppress the dc sources to obtain the circuit that
 - can be used to perform any required signal analysis



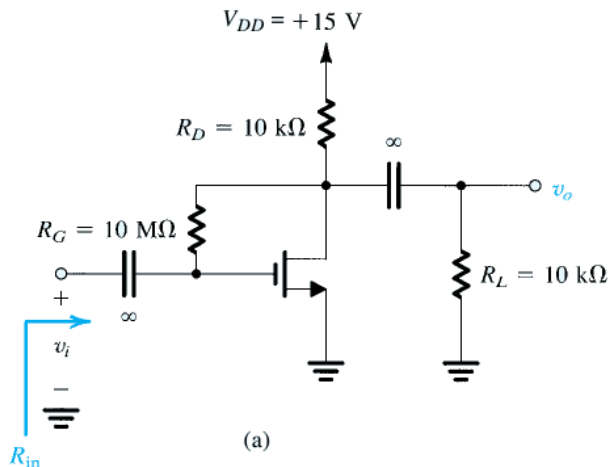
- the channel-length modulation effect can be
 - incorporated into the hybrid- π small-signal model
 - by placing the resistance r_o between drain and source
 - recall that $r_o = \frac{|V_A|}{I_D}$
 - where $V_A = \frac{1}{\lambda}$
 - recall that for a given process technology, $V_A \propto L$
 - I_D is value of dc drain current without the channel-length modulation taken into account
 - i.e. $I_D = \frac{1}{2}k_n V_{OV}^2 = \frac{1}{2}k_n' \frac{W}{L} (V_{GS} - V_t)^2$



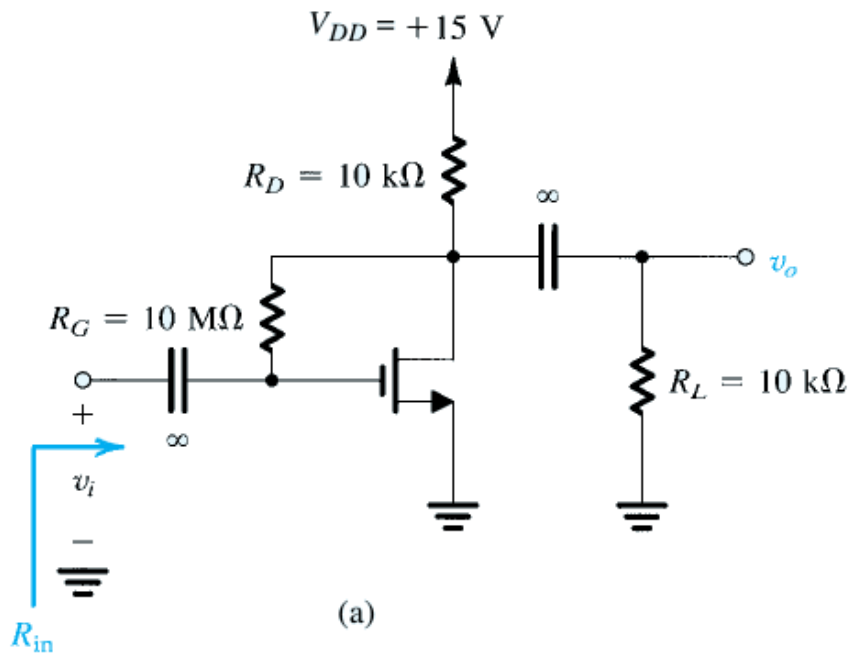
- Note that the small-signal model parameters g_m and r_o
 - depend on the dc bias point of the MOSFET
 - $g_m = k_n V_{OV} = \frac{2I_D}{V_{OV}} = \sqrt{2k_n'} \sqrt{\frac{W}{L}} \sqrt{I_D}$
 - $r_o = \frac{|V_A|}{I_D}$

**Example 5.10**

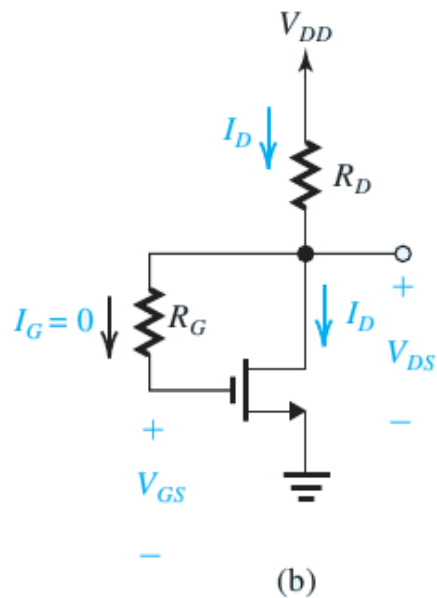
- Figure 5.39(a) shows a discrete common-source MOSFET amplifier utilizing a drain-to-gate resistance R_G for biasing purposes. Such a biasing arrangement will be studied in Section 5.7. The input signal v_i is coupled to the gate via a large capacitor, and the output signal at the drain is coupled to the load resistance R_L via another large capacitor. We wish to analyze this amplifier circuit to determine its small-signal voltage gain, its input resistance, and the largest allowable input signal. The transistor has $V_t = 1.5\text{V}$, $k_n'(W/L) = 0.25\text{mA/V}^2$, and $V_A = 50\text{V}$. Assume the coupling capacitors to be sufficiently large so as to act as short circuits at the signal frequencies of interest.

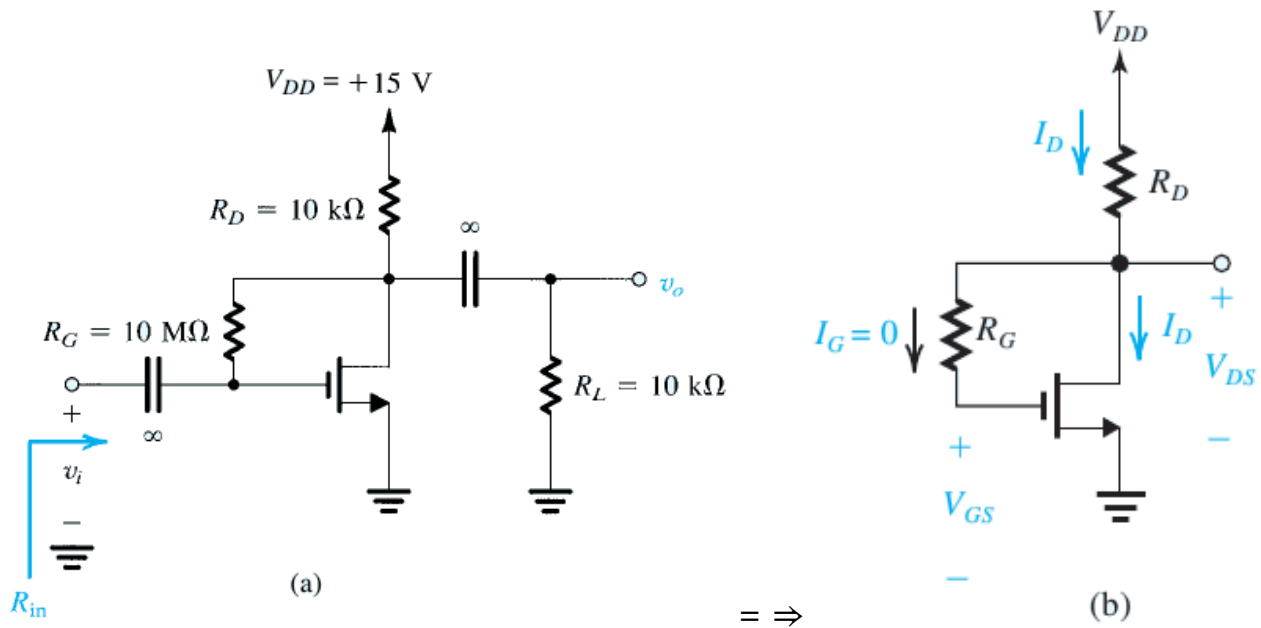


- Solution:
- $V_t = 1.5\text{V}$, $k_n'(W/L) = 0.25\text{mA/V}^2$, and $V_A = 50\text{V}$



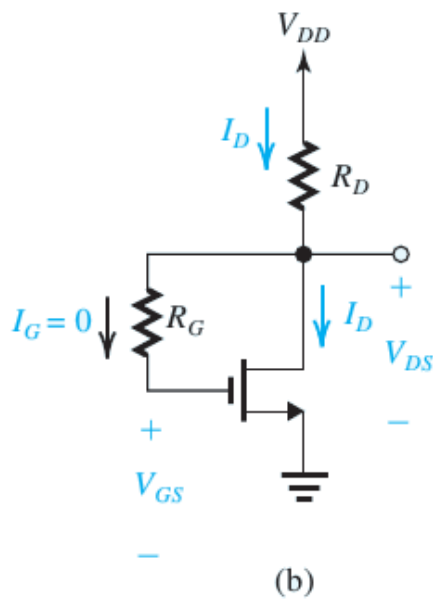
- first determine the dc operating point Q
- for this dc analysis
 - suppress ac signal sources
 - replace the two coupling capacitors by open circuits
 - (as these capacitors block dc currents)





Example 5.10, DC Analysis

- here $I_G = 0 \Rightarrow I_G R_G = 0$
 - $\Rightarrow V_G = V_D$
 - or $V_{GS} = V_{DS}$ as source is grounded
 - by KVL
 - $V_{DD} = I_D R_D + V_{DS}$
 - $\Rightarrow V_{GS} = V_{DS} = V_{DD} - I_D R_D = 15 - I_D(10k)$
 - for saturation
 - $V_{DS} \geq V_{GS} - V_t$
 - $V_{DS} - V_{GS} \geq -V_t$
 - $0 \geq -V_t \because V_{GS} = V_{DS}$
 - $0 \geq -1.5$ which is true $\Rightarrow$ MOSFET is in saturation



- neglecting the channel-length modulation effect on dc operating point

$$\circ \Rightarrow I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2$$

$$\blacksquare I_D = \frac{1}{2} (0.25m)(V_{GS} - 1.5)^2$$

$$\blacksquare I_D = 0.125m(V_{GS} - 1.5)^2$$

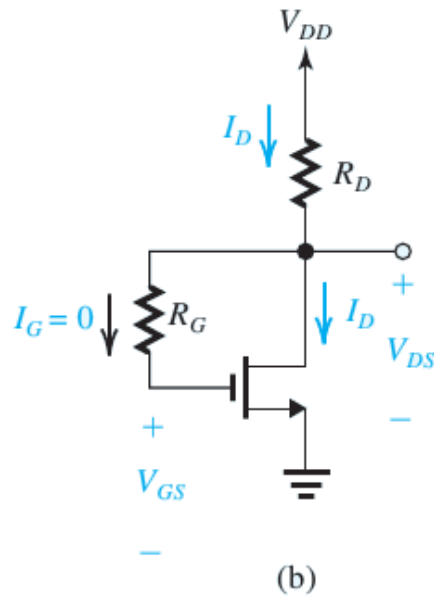
$$\blacksquare \text{As } V_{GS} = 15 - I_D(10k)$$

$$\blacksquare \Rightarrow I_D = \frac{15 - V_{GS}}{10k}$$

$$\blacksquare \text{thus } I_D = \frac{15 - V_{GS}}{10k} = 0.125m(V_{GS} - 1.5)^2$$

$$\blacksquare \frac{15 - V_{GS}}{10k} = 0.125m(V_{GS} - 1.5)^2$$

$$\blacksquare 15 - V_{GS} = 1.25(V_{GS} - 1.5)^2$$



$$\circ 15 - V_{GS} = 1.25(V_{GS}^2 + 2.25 - 3V_{GS})$$

- $15 - V_{GS} = 1.25(V_{GS}^2 + 2.25 - 3V_{GS})$

$$\circ 15 - V_{GS} = 1.25V_{GS}^2 + 2.8125 - 3.75V_{GS}$$

$$\blacksquare 1.25V_{GS}^2 - 2.75V_{GS} - 12.188 = 0$$

$$\blacksquare V_{GS} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\blacksquare V_{GS} = \frac{2.75 \pm \sqrt{(2.75)^2 - 4(1.25)(-12.188)}}{2(1.25)}$$

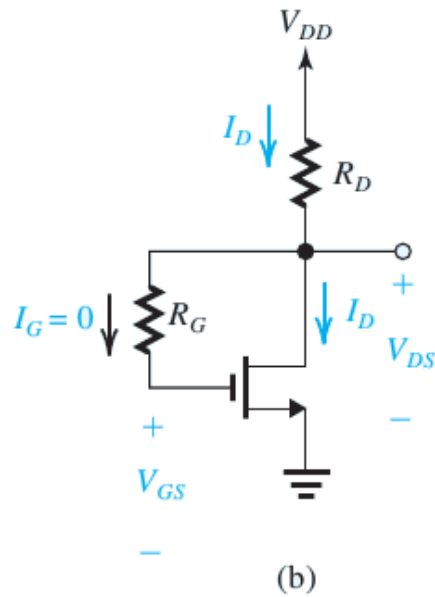
$$\blacksquare V_{GS} = \frac{2.75 \pm 8.2766}{2.5} = 4.4106, -2.2106$$

■ -ve V_{GS} is not valid as the MOSFET is in saturation region

$$\blacksquare \Rightarrow V_{GS} = 4.4106V$$

$$\blacksquare V_{DS} = V_{GS} = 4.4106V$$

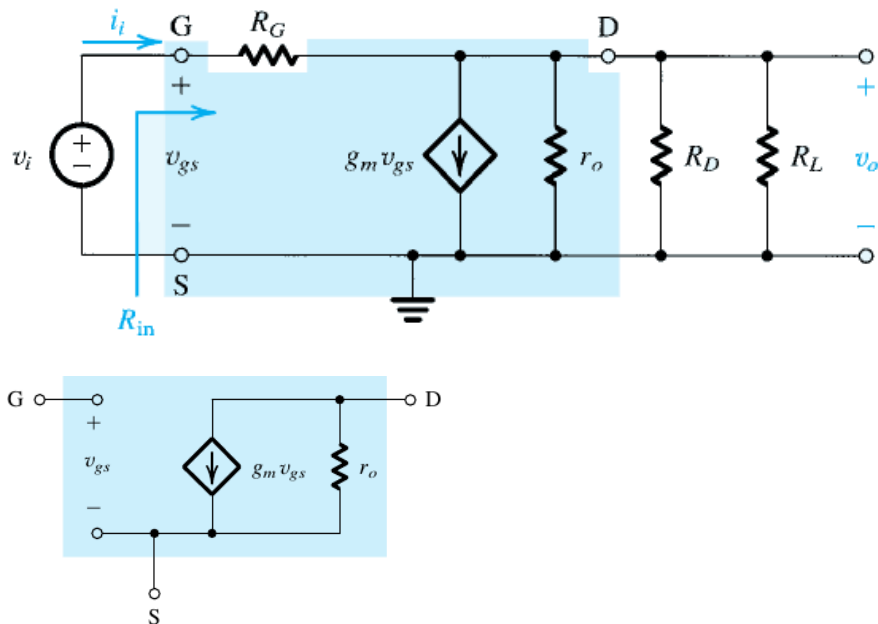
$$\blacksquare I_D = \frac{15 - V_{GS}}{10k} = \frac{15 - 4.4106}{10k} = 1.0589mA$$

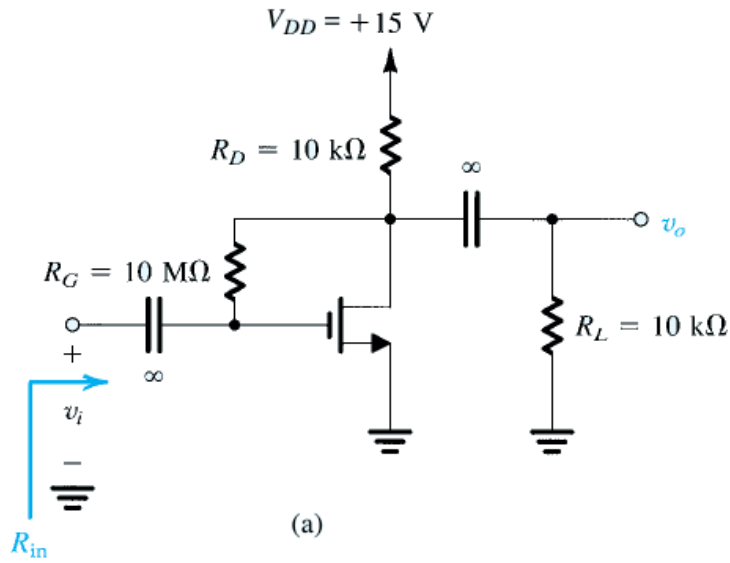


◦ $V_{OV} = V_{GS} - V_t = 4.4106 - 1.5 = 2.9106V$

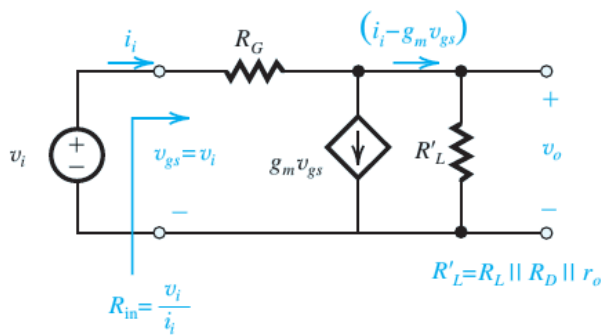
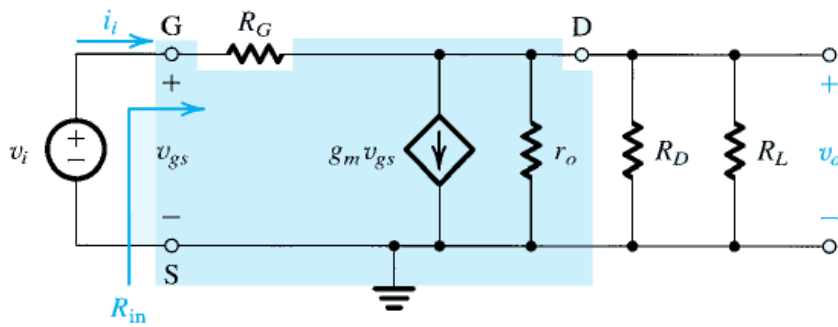
Example 5.10, Small-signal Analysis

- Small-signal Analysis
 - Replace the MOSFET with its small-signal model
 - suppress the DC sources (replace V_{DD} by short-circuit)
 - replace coupling capacitors with short circuits

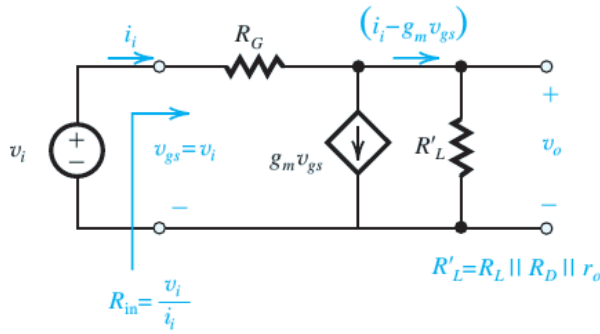




- as r_o , R_L and R_D are in parallel
 $\circ \Rightarrow R_L' = r_o \parallel R_L \parallel R_D$

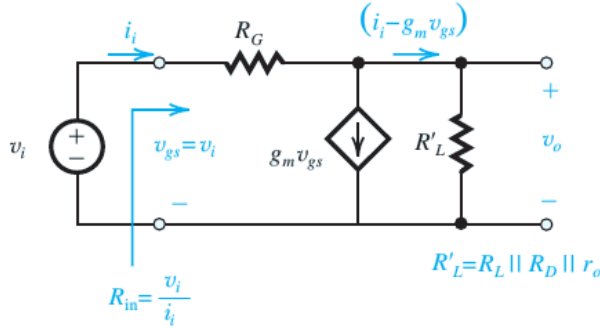


- $R_L' = r_o \parallel R_L \parallel R_D$



- $r_o = \frac{V_A}{I_D} = \frac{50}{1.0589m} = 47.219k\Omega$
- $R'_L = r_o \parallel R_L \parallel R_D = \left(\frac{1}{r_o} + \frac{1}{R_L} + \frac{1}{R_D} \right)^{-1} = \left(\frac{1}{47.219k} + \frac{1}{10k} + \frac{1}{10k} \right)^{-1} = 4.52k\Omega$
- $g_m = k_n V_{OV} = (0.25m)(2.9106) = 0.72765m \frac{A}{V}$
- input resistance, $R_{in} = \frac{v_i}{i_i}$

- input resistance, $R_{in} = \frac{v_i}{i_i}$



- by KCL, $i_i = g_m v_{gs} + i_{R'_L} = g_m v_{gs} + \frac{v_o}{R'_L} = g_m v_{gs} + \frac{v_o}{4.52k}$
- ohms' law at $R_G \Rightarrow i_i R_G = v_i - v_o$
 - $\Rightarrow i_i = \frac{v_i - v_o}{R_G}$
- thus $i_i = \frac{v_i - v_o}{R_G} = g_m v_{gs} + \frac{v_o}{4.52k}$
 - $\frac{v_{gs} - v_o}{R_G} = g_m v_{gs} + \frac{v_o}{4.52k}$
- $\frac{v_{gs} - v_o}{R_G} = g_m v_{gs} + \frac{v_o}{4.52k}$
 - $\frac{v_{gs}}{R_G} - \frac{v_o}{R_G} = g_m v_{gs} + \frac{v_o}{4.52k}$
 - $\frac{v_{gs}}{R_G} - g_m v_{gs} = \frac{v_o}{4.52k} + \frac{v_o}{R_G}$
 - $v_{gs} \left(\frac{1}{R_G} - g_m \right) = v_o \left(\frac{1}{4.52k} + \frac{1}{R_G} \right)$
 - $A_v = \frac{v_o}{v_i} = \frac{v_o}{v_{gs}} = \left(\frac{1}{R_G} - g_m \right) / \left(\frac{1}{4.52k} + \frac{1}{R_G} \right)$
 - $A_v = \left(\frac{1}{10M} - 0.728m \right) / \left(\frac{1}{4.52k} + \frac{1}{10M} \right) = -3.2741 \frac{V}{V}$
 - As $A_v = \frac{v_o}{v_i} = -3.2741 \frac{V}{V} \Rightarrow v_o = A_v v_i = A_v v_{gs}$
 - as $R_{in} = \frac{v_i}{i_i} = \frac{v_{gs}}{i_i}$

$$\blacksquare \text{ where } i_i = \frac{v_i - v_o}{R_G}$$

$$\circ \Rightarrow R_{in} = \frac{v_{gs}}{i_i} = \frac{R_G v_{gs}}{v_i - v_o} = \frac{R_G v_{gs}}{v_{gs} - A_v v_{gs}} = \frac{R_G v_{gs}}{v_{gs}(1 - A_v)} = \frac{R_G}{1 - A_v}$$

$$\circ R_{in} = \frac{R_G}{1 - A_v} = \frac{10M}{1 - (-3.2741)} = 2.3397M\Omega$$

- to avoid cutoff region

- $v_{GS} \geq V_t$
- $v_{GS_{min}} \geq V_t$
- $V_{GS} - \hat{v}_{gs} \geq V_t$
- $4.4106 - \hat{v}_{gs} \geq 1.5$
- $-\hat{v}_{gs} \geq 1.5 - 4.4106$
- $-\hat{v}_{gs} \geq -2.9106$
- $\hat{v}_{gs} \leq 2.9106$

- to avoid triode region

- $v_{DS} \geq v_{GS} - V_t$
- $v_{DS_{min}} \geq v_{GS_{max}} - V_t$
- $V_{DS} - \hat{v}_{ds} \geq (V_{GS} + \hat{v}_{gs}) - V_t$
- as $A_v = \frac{v_o}{v_i} = \frac{v_{ds}}{v_{gs}} \Rightarrow \hat{v}_{ds} = |A_v| \hat{v}_{gs}$
- $V_{DS} - |A_v| \hat{v}_{gs} \geq V_{GS} + \hat{v}_{gs} - V_t$
- as $V_{DS} = V_{GS} = 4.4106V$, $V_t = 1.5V$
- $-|A_v| \hat{v}_{gs} \geq \hat{v}_{gs} - V_t$
- $-3.2741 \hat{v}_{gs} \geq \hat{v}_{gs} - 1.5$

- thus to avoid cutoff region

- $\hat{v}_{gs} \leq 2.9106$

- thus to avoid triode region

- $1.5 \geq \hat{v}_{gs} + 3.2741 \hat{v}_{gs}$
 - $1.5 \geq 4.2741 \hat{v}_{gs}$
 - $\hat{v}_{gs} \leq \frac{1.5}{4.2741} = 0.35095V$
-